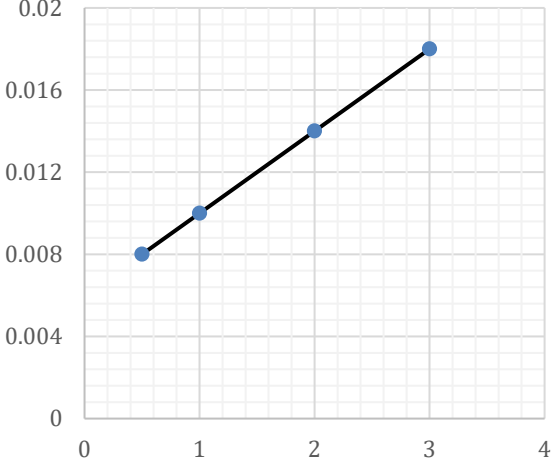


Question Number	Answer	Additional Guidance	Mark
2(a)(i)	peptide (bond)		Grad (1)

Question Number	Answer	Additional Guidance	Mark
2(a)(ii)	hydrolysis		Grad (1)

Question Number	Answer	Additional Guidance	Mark
2(b)(i)	0.018		Expert (1)

Question Number	Answer	Additional Guidance	Mark
2(b)(ii)	A graph showing the following features: • A axes correct (x – substrate concentration / y – initial rate of reaction / (1) • L axes correctly labelled and with units (1) • S suitable scale on the x-axis (1) • P correct plotting on a linear scale on y (1)	initial rate of reaction / au min⁻¹  substrate concentration (%) Ignore line Allow percentage concentration for y Allow ecf if miss mp1 for mp3 and 4 Allow use of standard form on y axis Allow ecf from 2bi in the plot including if 2bi blank	Expert (4)

Question Number	Answer	Additional Guidance	Mark
2(b)(iii)	1.8		Clerical (1)

Question Number	Answer	Additional Guidance	Mark
2(b)(iv)	An explanation that includes the following points: - the rate increases (linearly) as the substrate concentration increases (1) - because there (is more substrate so) {more collisions / more ES} complexes (1) - does not level off because the {enzyme concentration does not become limiting / enzyme's active sites are not all occupied at the highest substrate concentration} (1)	Accept positive correlation Reverse argument if their graph less steep or plateau after 2% If graph goes down at 3% ignore	expert (3)

Question Number	Answer	Additional Guidance			Mark															
2(c)(i)	A table drawn with the following features: • suitable table drawn (1) • headings with units (1) • all data correctly entered (1)	<table><tr><td>pH</td><td>enzyme activity in stem / a.u.</td><td>enzyme activity in fruit / a.u.</td></tr><tr><td>4.(0)</td><td>1.8</td><td>0.7</td></tr><tr><td>7.(0)</td><td>5.1</td><td>2.3</td></tr><tr><td>8.5</td><td>6.6</td><td>2.5</td></tr><tr><td>10.(0)</td><td>1.9</td><td>1.5</td></tr></table>	pH	enzyme activity in stem / a.u.	enzyme activity in fruit / a.u.	4.(0)	1.8	0.7	7.(0)	5.1	2.3	8.5	6.6	2.5	10.(0)	1.9	1.5	Allow ±0.1 for each cell		expert (3)
pH	enzyme activity in stem / a.u.	enzyme activity in fruit / a.u.																		
4.(0)	1.8	0.7																		
7.(0)	5.1	2.3																		
8.5	6.6	2.5																		
10.(0)	1.9	1.5																		

Question Number	Answer	Additional Guidance			Mark
2(c)(ii)	An answer that includes the following points: - stem (extract) because {more active / higher} (than fruit extract) (1) - at all pH values (1) - manipulation of data to support mp 1 (1)	Accept reverse Accept ecf if say fruit for mp 1 e.g. stem {4.1 au / 2.64 times} more active at pH 8.5 / comparison of a pair or more of values Accept ecf if say fruit for mp 1			expert (3)